Introduction to Computing

ASSIGNMENT 4: REGULAR EXPRESSIONS & DFAS

Instructions: For each of the scenarios below, you are given an alphabet and a language description. You must design a system that correctly accepts or rejects strings based on the rules. For every question, complete Tasks A, B, and C.

Question 1: Starting Character

Consider the alphabet $\Sigma = \{0, 1\}$. The system accepts all strings that begin with the character '1'. Any characters that follow the initial '1' do not matter, and a single '1' is also acceptable.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram (indicating the start state, transitions, and accept states).
- **Task C:** Provide two strings that the machine will **accept**, and two strings it will **reject**.

Question 2: Ending Substring

Consider the alphabet $\Sigma = \{a, b\}$. The system accepts all strings that end specifically with the sequence "ab". The string must contain "ab" at the very end to be accepted.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 3: Containing a Substring

Consider the alphabet $\Sigma = \{0, 1\}$. The system accepts any string that contains the substring "00" anywhere within it. It can be at the beginning, middle, or end.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 4: Exact Count

Consider the alphabet $\Sigma = \{a, b\}$. The system accepts all strings that contain exactly one 'a'. The string can contain any number of 'b's before or after this single 'a', but if there are zero 'a's or more than one 'a', it must be rejected.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 5: Even Length

Consider the alphabet $\Sigma = \{0, 1\}$. The system accepts all strings that have an even length (i.e., a total of 0, 2, 4, 6... characters). Note that the empty string (represented by ε or λ) has a length of 0, which is even.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 6: Exact Match

Consider the alphabet $\Sigma = \{0, 1\}$. The system is a simple password checker that accepts *only* the exact string "101". Any string that is longer, shorter, or contains different characters must be rejected.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 7: Start and End Requirements

Consider the alphabet $\Sigma = \{x, y\}$. The system accepts all strings that start with an 'x' AND end with a 'y'. (Note: The string "xy" is the shortest string accepted).

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 8: Homogenous Strings

Consider the alphabet $\Sigma = \{0, 1\}$. The system accepts strings that are made up entirely of '1's. This includes a single '1', "11", "111", and also the empty string. If a '0' appears anywhere in the string, it is rejected.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 9: Minimum Length

Consider the alphabet $\Sigma = \{a, b\}$. The system accepts all strings that have a length of at least 2. The actual characters do not matter, as long as there are 2 or more of them.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.

Question 10: The "OR" Condition

Consider the alphabet $\Sigma = \{0, 1\}$. The system accepts any string that either starts with a '0' OR ends with a '1'. If a string fulfills either (or both) of these conditions, it is accepted.

- **Task A:** Write the Regular Expression.
- **Task B:** Draw the DFA state diagram.
- **Task C:** Provide two strings that the machine will accept, and two strings it will reject.